

Geriatric Nephrology

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Aging is a natural and progressive process where there is a slow decline in cellular function that ultimately may become manifest as changes in overall organ function. These inevitable changes that occur with aging need to be distinguished from diseases that are more common among older people, such as heart failure, hypertension, and diabetes mellitus (DM), all of which usually have more dramatic effects on organ function. Although it is often hard to distinguish age-related from disease-mediated changes in organ function, it is generally true that they can act synergistically to alter the rate of functional decline of the kidney and to make the kidney more susceptible to injury and less likely to fully recover. In addition, the age-related decline in kidney function has important implications for medication dosing.

AGING-ASSOCIATED CHANGES IN THE KIDNEY

Anatomic Changes

The human kidney reaches a maximum size of approximately 400 g (12 cm in length) in the fourth decade of life. A natural decline of approximately 10% in kidney mass per decade then follows. This natural decline is associated with cortical thinning and a decrease in the number of functional nephrons. An analysis of kidney tissue from healthy living donors examined the changes in nephron number and structure over time.¹ The number of nonsclerotic glomeruli was 48% lower in the oldest age groups (70–75 years) compared with those aged 18 to 29 years. In contrast, cortical volume was only 16% lower and the proportion of globally sclerotic glomeruli was only 15% higher. This incomplete representation of nephron loss with aging by either increased glomerulosclerosis or cortical volume decline is consistent with atrophy and reabsorption of globally sclerotic glomeruli and hypertrophy of remaining nephrons.

Other anatomic changes with aging include an increase in simple parenchymal and parapelvic kidney cysts and in angiomyolipomas.²

Glomerular Changes

Structural glomerular changes with age include basement membrane thickening and development of focal or global glomerulosclerosis, which increases to 10% to 30% and in some studies even exceeds 70% of glomeruli by the eighth decade (Figs. 93.1 [unchanged] and 93.2 [unchanged]).³ The cross-sectional area of the tuft among preserved glomeruli in older people is often larger than average, consistent with glomerular hypertrophy. Neither kidney function nor chronic kidney disease (CKD) risk factors explain the strong association between age and glomerulosclerosis in healthy adults.

Tubular and Interstitial Changes

Tubulointerstitial injury associated with aging is most pronounced in the outer medulla, with tubular dilation and atrophy, mononuclear cell infiltration, and interstitial fibrosis. Some tubules (especially in the distal tubule and collecting duct) may develop small diverticuli; these diverticuli may play a role in the development of upper urinary tract infections (pyelonephritis) by harboring bacteria.⁴

Vascular Changes

Arterioles often develop hyalinosis with aging. Thickening of the arterioles with an increase in the ratio of medial thickness to lumen diameter is common with aging but is observed almost exclusively in hypertensive individuals.⁴ The arcuate arteries become more angulated and irregular with aging, and there is increased tortuosity and spiraling of the interlobar vessels. These changes occur independently of hypertension but are augmented in its presence. With aging, some afferent arterioles, particularly of juxtamedullary glomeruli, develop vascular shunts to the efferent arterioles, thereby bypassing glomeruli, leading to “aglomerular arterioles.”⁵

Nephrosclerosis

Thus, the aging kidney develops some combination of glomerulosclerosis, tubular atrophy, interstitial fibrosis, and arteriosclerosis. In one study, the presence of two or more of these abnormalities was found in 2.7% in young kidney donors and increased to 73% in the oldest group of donors.⁶

AGING-ASSOCIATED CHANGES IN KIDNEY FUNCTION

Glomerular Filtration Rate

Inulin clearance studies document a progressive fall in glomerular filtration rate (GFR) after the age of 40 years, with a relatively greater decline in males (Fig. 93.3 [unchanged]).⁷ However, the fall in GFR does not appear to be inevitable; in as many as one-third of patients who remain normotensive, there is no decrease in creatinine clearance with age.² Studies of potential kidney donors show that the rate of decline of GFR is between 6.3 and 7.5 mL/min per decade.⁸ Of note, the GFR assesses the function of all nephrons, and the single-nephron GFR assesses the function of individual nephrons. Single-nephron GFR does not vary significantly according to age (among kidney donors <70 years).⁹ Thus, the fall in GFR is likely because the number of nephrons declines with age.¹⁰

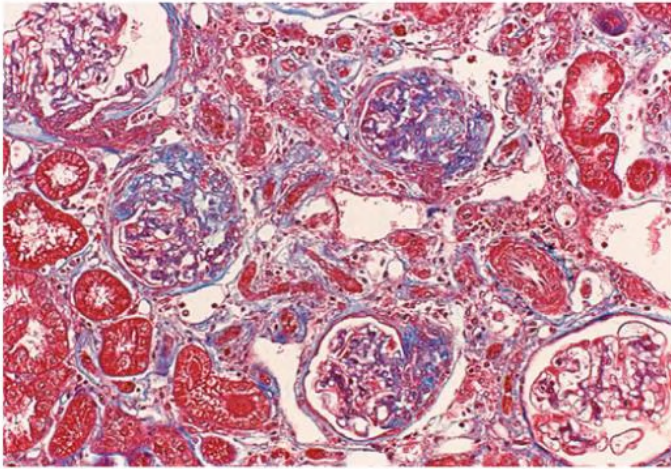


Fig. 93.1 Glomerulosclerosis and Tubulointerstitial Fibrosis in an Aging Rat. Similar changes, consisting of focal segmental glomerulosclerosis, tubular atrophy, and interstitial fibrosis, occur in humans. (Trichrome stain; original magnification $\times 400$.)

Age-Specific Incidence of Glomerulosclerosis

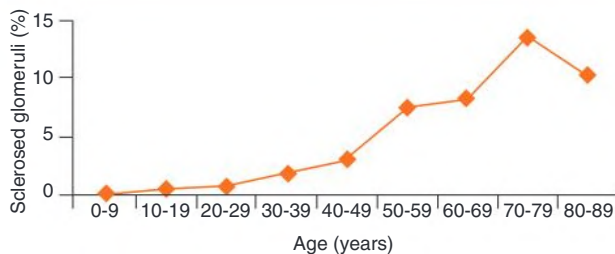


Fig. 93.2 The incidence of glomerulosclerosis increases with aging. (Modified from Rule AD, Sasiwimonphan K, Lieske JC, et al. Characteristics of renal cystic and solid lesions based on contrast-enhanced computed tomography of potential kidney donors. *Am J Kidney Dis.* 2012;59[5]:611–618.)

In addition to the decrease in GFR with aging, there may be a reduction in kidney “reserve.” Although some studies suggest that aging humans show a normal increase in GFR after amino acid infusion, others have shown a marked reduction in increases in kidney plasma flow (KPF) and GFR in response to concurrent infusion of amino acids and dopamine in healthy older individuals.^{11,12}

Kidney Plasma Flow

KPF also decreases from a mean of 650 mL/min in the fourth decade to 290 mL/min by the ninth decade, with increasing kidney vascular resistance (Fig. 93.4).⁴ Because KPF decreases relatively more than GFR, filtration fraction (defined as GFR/KPF) increases with age. Studies have demonstrated that there is a true reduction in kidney blood flow when it is factored for kidney mass.¹³ The decrease in kidney blood flow especially involves the cortex, and blood flow to the medulla is relatively preserved.

Albuminuria

The prevalence of albuminuria increases progressively after the age of 40 years. The increased prevalence is most marked in diabetic and hypertensive patients but is also observed in patients lacking these risk factors. However, whether age per se is associated with proteinuria is debatable.¹⁴

Hematuria

Malignancies of the urinary tract are more common in older patients, and so the diagnostic workup for hematuria in patients aged 50 years

GFR Decreases With Age

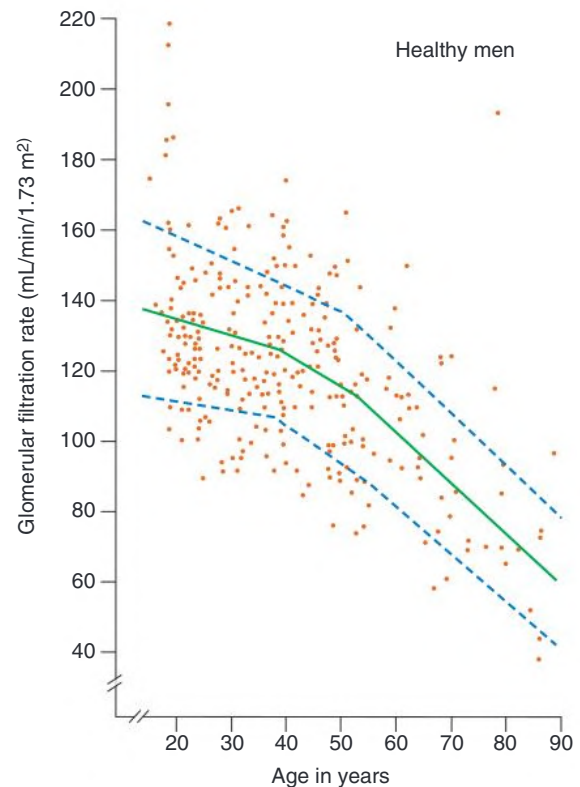
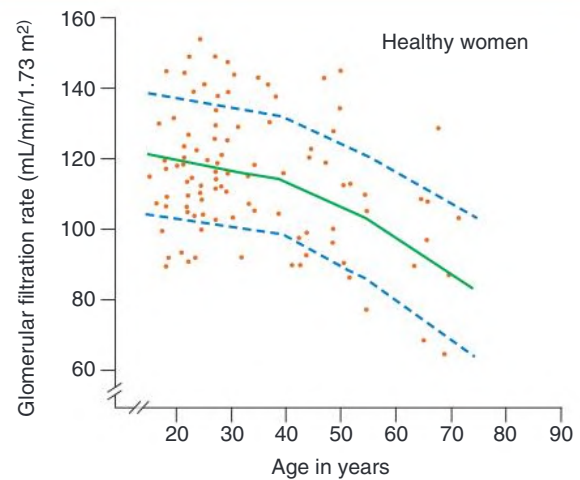


Fig. 93.3 Glomerular Filtration Rate (GFR) Decreases With Age. GFR (inulin clearance) begins to fall at age 40 years, and the rate of decline is more rapid in males than in females. (Modified from Takazakura E, Sawabu N, Handa A, et al. Intrarenal vascular changes with age and disease. *Kidney Int.* 1972;2:224–230.)

should include cystoscopy and urinary tract imaging. Renal cell carcinomas (RCCs) are most commonly diagnosed after the fifth decade of life. RCCs may be more aggressive in the elderly, and treatment decisions can be difficult, especially in the setting of significant CKD in which nephrectomy may lead to dialysis dependence.¹⁵ However, increased use of biopsy to determine whether a mass lesion is cancer and localized resection of the mass alone or surveillance over time have become more common and preserve kidney function. Other causes of hematuria may need to be considered, as discussed in Chapters 4 and 63.

Plasma Flow Decreases With Age

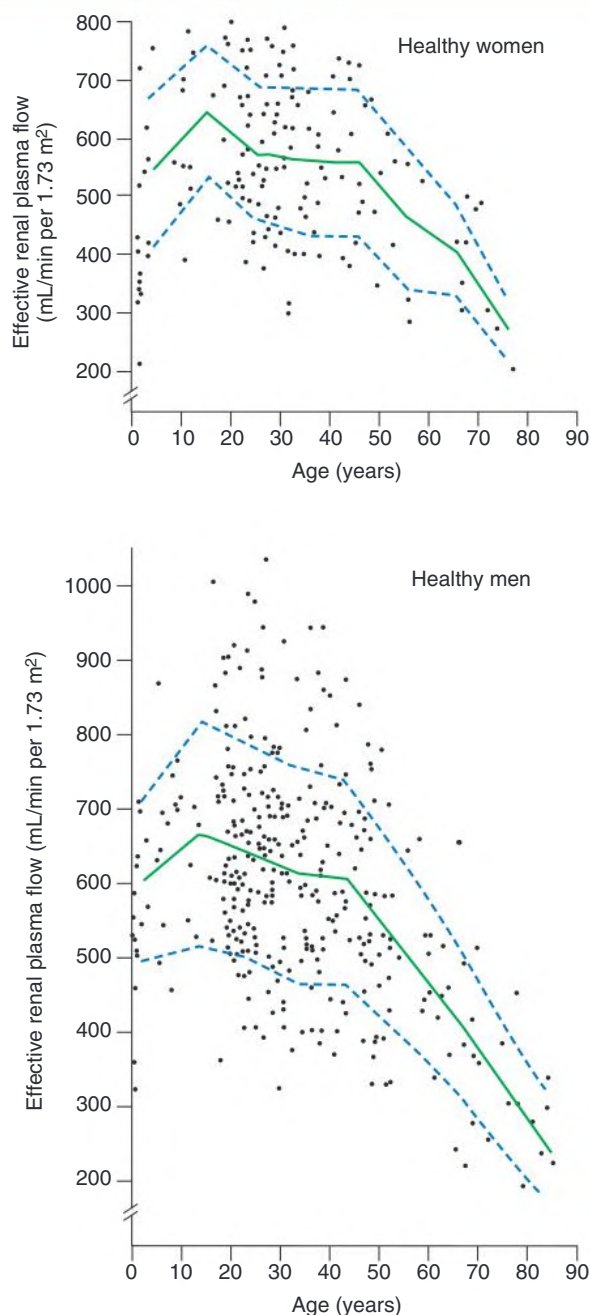


Fig. 93.4 Renal Plasma Flow (RPF) Decreases With Age. RPF (ρ -aminohippurate clearance) begins to fall rapidly after the age of 50 years, and the rate of decline is more rapid in males than in females. (Modified from Takazakura E, Sawabu N, Handa A, et al. Intrarenal vascular changes with age and disease. *Kidney Int.* 1972;2:224–230.)

ASSESSMENT OF GLOMERULAR FILTRATION RATE IN THE ELDERLY

Serum creatinine becomes a less reliable indicator of GFR after aging because after the age of 60 years, there is a progressive decrease in urinary creatinine excretion, which largely reflects that muscle mass decreases with aging (Fig. 93.5).¹⁶ There is no consensus on the optimal approach to estimation of GFR in elderly people. Although the Modification of Diet in Renal Disease (MDRD) study equation and

Urinary Creatinine Excretion as a Function of Age

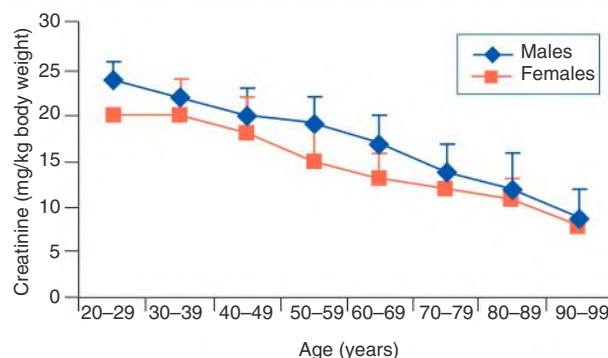


Fig. 93.5 Urinary creatinine excretion (adjusted for body weight) decreases with age. (Modified from Denic A, Mathew J, Lerman LO et al. Single-nephron glomerular filtration rate in healthy adults. *New Engl J Med.* 2017;376[24]:2349–2357.)

the Cockcroft-Gault formula for estimating GFR use age in their calculations (see Chapter 3), neither has been validated in people older than 70 years and both underestimate true GFR in those older than 65 years compared with gold standard techniques. Although the MDRD equation may be more accurate than the Cockcroft-Gault formula,¹⁷ serum cystatin C, which is independent of muscle mass, may be superior to both.¹⁸ The Berlin Initiative Study equation for estimating GFR was derived from patients who were older than 70 years using iohexol clearance as the gold standard and is particularly accurate for classifying patients whose true GFR is between 30 and 90 mL/min/1.73m².¹⁹ Another equation was recently developed and has improved validity across the full age spectrum, with less bias at the extremes of age and perhaps allowing for better clinical decision making.²⁰

The implementation of routine estimated GFR (eGFR) reporting has led to a higher number of older adults diagnosed with CKD, leading to increased referrals to nephrologists and increased anxiety among patients.⁹ As discussed later, the clinical significance of this diagnosis (which is largely driven by age in estimation equations) is not clear.

PREVALENCE OF CHRONIC KIDNEY DISEASE IN THE ELDERLY

According to the U.S. Renal Data System's Annual Data Report, the prevalence of CKD defined by eGFR is increasing in people aged 65 years and older with nearly 25% of those older than 85 years showing some level of CKD (Fig. 93.6).²¹ Of note, higher increases in the apparent overall prevalence of CKD occurred around 2005 to 2006 and 2015 to 2016 and are likely attributable to the introduction of stage-specific CKD codes and the shift from the International Classification of Diseases, Ninth Revision, Clinical Modification (ICD-9-CM) to the ICD-10-CM classification system, respectively.

eGFR is attractive for classifying the severity of CKD because eGFR is an independent predictor of mortality and other adverse events. However, this may not be true in patients older than 65 years of age. For instance, eGFR of 50 to 59 mL/min/1.73 m² may not be associated with increased mortality among patients age 65 years or older compared with patients with eGFR of more than 60 mL/min/1.73 m².²² Additionally, cardiovascular mortality begins to increase in those older than 65 years with eGFR less than 45 mL/min/1.73 m².²³ However, other studies have shown that both low eGFR and high albuminuria were independently associated with mortality and end-stage kidney disease (ESKD) regardless of age across a wide range of populations.²⁴

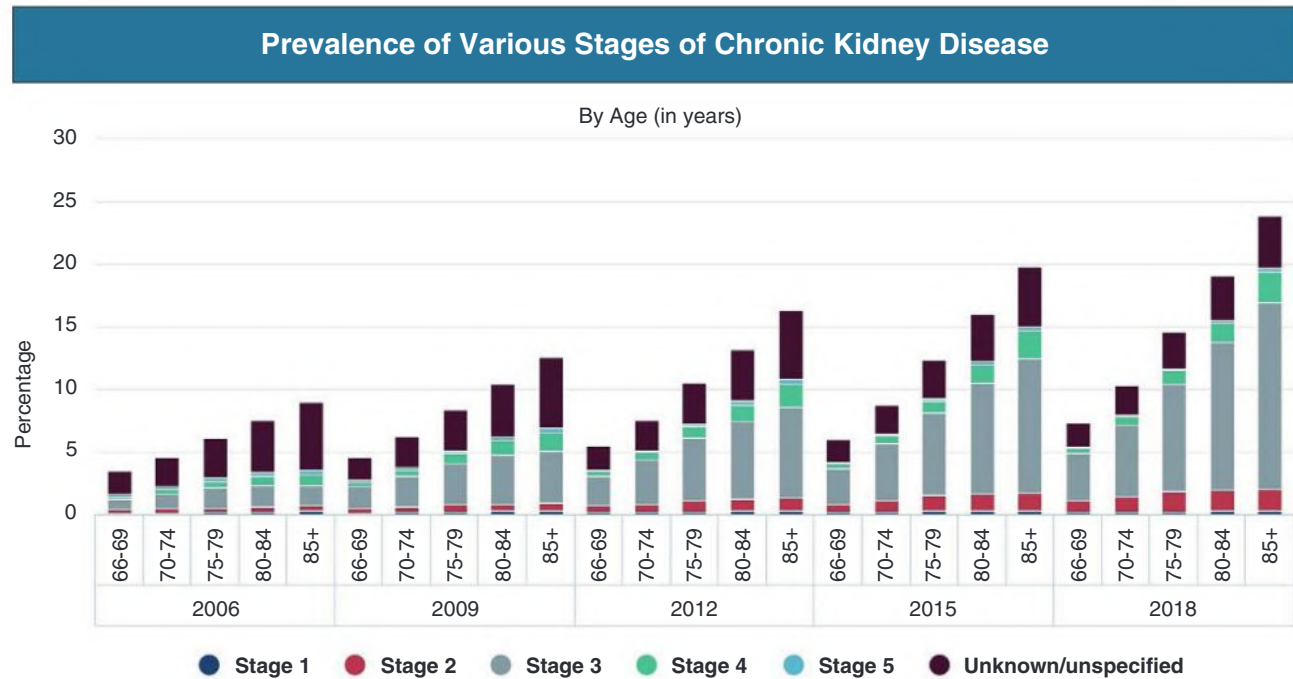


Fig. 93.6 Prevalence of various stages of chronic kidney disease (CKD) in people aged >65 years over 2006 to 2018 demonstrating increases in stage 3 to 5 CKD over time in this population. (Data from U.S. Renal Data System. www.usrds.org.)

The interpretation of small decreases in eGFR in those older than 65 years has led to debate and whether the term *chronic kidney disease* in such cases should be replaced with *age-related reduced kidney function*.²⁵

RISK FACTORS FOR CHRONIC KIDNEY DISEASE IN THE ELDERLY

The variability in the severity of aging-related kidney disease in humans has suggested that other factors besides aging per se may contribute to the loss of GFR that is seen in older populations. A recent study found that in a cohort of patients older than 65 years with stage 3A to 5 CKD, up to 60% of patients showed continued declines in kidney function over 2 years.²⁶ Factors associated with progressive loss of eGFR in prospective and retrospective studies include use of oral analgesics, metabolic syndrome, hyperuricemia, proteinuria, DM, and baseline CKD.²⁶ In another prospective study, increased levels of physical activity correlated with a lower risk for rapid GFR decline (defined as loss > 3 mL/min/1.73 m² per year, based on cystatin C levels) in a general population of older adults.²⁷

PATHOGENESIS OF AGE-RELATED CHRONIC KIDNEY DISEASE

A variety of mechanisms have been proposed for the aging-related kidney changes that are reflected in declining GFR, decreased tubular function, progressive fibrosis, and vascular changes (Box 93.1). The kidney is an important source of the antiaging hormone klotho. Klotho is a transmembrane protein and a coreceptor for fibroblast growth factor-23 (FGF-23). Klotho has a role in modulating many aging-related pathways associated with cellular senescence and regeneration, and mice with klotho deficiency recapitulate features of systemic and kidney aging.²⁸ As klotho levels fall during aging, signaling through the Wnt pathway increases, leading to fibrosis and vascular calcification.²⁹ Klotho (and its deficiency) may be a future target for antiaging strategies.

BOX 93.1 Proposed Mechanisms for Aging-Associated Kidney Disease

Hemodynamic

- Glomerular hypertension and hyperfiltration
- Intrarenal activation of renin-angiotensin system
- Endothelial dysfunction (loss of nitric oxide)
- Kidney ischemia
- Decreased renalase

Metabolic

- Accumulation of advanced glycation end-products
- Chronic effects of uric acid
- Chronic metabolism of endogenous and exogenous fructose

Cellular Dysfunction

- Oxidative stress
- Decreased klotho
- Senescence (with telomere shortening and loss of mitochondria)

Additional mechanisms of kidney aging may involve telomere shortening of chromosomal DNA, declining levels of peroxisome proliferator-activated receptor- γ , loss of mitochondria, and accelerated apoptosis. Aging-associated kidney disease also may be mediated by activation of the renin-angiotensin system, which may lower kidney klotho expression. Other mechanisms may include hyperfiltration injury and glomerular hypertension as well as vascular changes associated with arteriolar stiffening, progressive reduction in nitric oxide production by endothelial cells, progressive capillary loss with ischemia, and accumulation of advanced glycation endproducts.^{29–35} The possibility that sugar intake and/or fructose metabolism may be involved has also been suggested by the finding that mice lacking the ability to metabolize fructose appear to be protected from age-associated kidney injury.³⁶

FLUID AND ELECTROLYTES IN AGING

Sodium Balance and Hypertension

Aging is associated with impaired excretion of a salt load and defective conservation in the setting of sodium restriction.³⁷ Proximal sodium reabsorption is increased in aging, whereas distal sodium reabsorption may be reduced.³⁸ Because the diet of most individuals in wealthier countries contains excess sodium (8–10 g of salt daily), there is a tendency for total body sodium excess among elderly people. This relative defect in sodium excretion and increased total body sodium may be predisposing factors for the development of hypertension, the prevalence of which increases with age. After the age of 60 years, most people are hypertensive (Fig. 93.7).³⁹ Salt sensitivity occurs in more than 85% of aging people, and sodium restriction will result in a significant fall (>10 mm Hg) in mean arterial pressure.⁴⁰ Populations that ingest low-sodium diets, such as the Yanomami Indians of southern Venezuela, do not show an increase in blood pressure (BP) with age.⁴¹ Loss of vascular compliance also may contribute to aging-associated hypertension, as may endothelial dysfunction, perhaps mediated by oxidative stress. Aging-associated kidney and vascular changes may explain why correction of secondary forms of hypertension (e.g., primary aldosteronism, Cushing syndrome, and renovascular hypertension) is less effective at curing hypertension in older patients. In one study, diastolic BP (DBP) fell to less than 90 mm Hg in 24 of 25 patients younger than 40 years after treatment of the mechanism responsible for the secondary hypertension but in only 38 of 61 patients older than 40 years.⁴² Recent data suggest that treatment of hypertension in the elderly (>75 years) should be similar to that in the younger population because treating to a systolic BP (SBP) target of less than 120 mm Hg (compared with an SBP target of <140 mm Hg) resulted in significantly lower rates of fatal and nonfatal major cardiovascular events.⁴³ Reaching these lower BP goals in the elderly may be more difficult, require more medications, and have risk for orthostatic and diastolic hypotension. Hypotension in the frail elderly patient may be associated with an increased risk of falls and morbidity.

Osmoregulation and Water Handling

The most common electrolyte abnormalities in the elderly are the consequence of impaired water handling with aging. Hyponatremia has been found in up to 11% of the ambulatory geriatric population

and 5.3% of hospitalized elderly patients.⁴⁴ Hypernatremia is found in about 1% of patients older than 60 years and admitted to the hospital.⁴⁴ Both concentration and dilution of the urine are affected by aging and account for part of the susceptibility to dysnatremias.

In the elderly, the maximal urinary osmolality and thirst response to hyperosmolality are reduced, which may predispose to dehydration and hypernatremia. The impairment in urine concentrating ability results from a defect in the concentrating gradient in the medullary region and can lead to nocturia.⁴⁵ Compounding the risk for hypernatremia is that the ill elderly patient may not have ready access to water.

The elderly also have an impaired ability to dilute the urine and thus have a decreased ability to excrete a water load, leading to an increased predisposition to hyponatremia that is often compounded by the use of medications such as thiazide diuretics and selective serotonin reuptake inhibitors.⁴⁶ Whether age per se is an independent risk factor for the development of hyponatremia has been questioned, because after adjustment for frailty, the relationship between age and sodium disorders is no longer significant.

Other Tubular Defects and Electrolyte Problems

Potassium excretion is impaired in the elderly, and the transtubular potassium gradient is decreased.⁴⁷ Hyperkalemia occurs more frequently in elderly patients treated with drugs that interfere with potassium excretion (such as potassium-sparing diuretics or agents that block the renin-angiotensin-aldosterone system). Other factors contributing to hyperkalemia in the elderly include decreased GFR, lower basal levels of aldosterone, and tubulointerstitial scarring. Hypokalemia is also common because of kidney (especially with thiazide diuretics) or extrarenal losses.

Most elderly individuals can maintain acid-base balance under normal conditions. However, during conditions of stress when acid production is increased (sepsis or acute kidney injury [AKI]), an inability to excrete an additional acid load may be uncovered. This is supported by a study demonstrating that elderly patients could not increase net acid excretion to the same level as younger adults in response to a protein meal.⁴⁸ This tendency to metabolic acidosis becomes more manifest in the presence of CKD.

Hypercalcemia occurs in 1% to 3% of elderly patients. Causes include malignancy, hyperparathyroidism, immobilization, and use of thiazide diuretics. Many patients with thiazide-associated

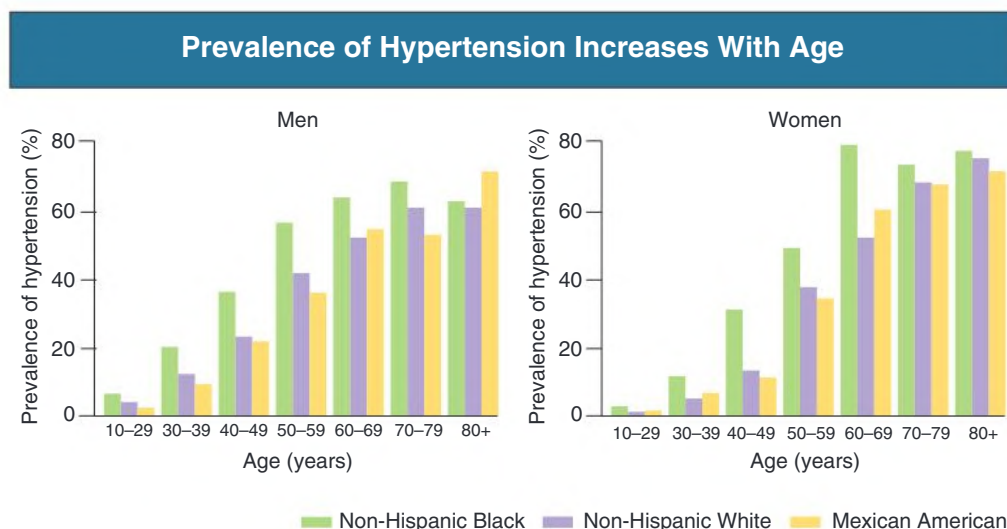


Fig. 93.7 Prevalence of hypertension based on age, sex, and race. (Modified from Burt VL, Whelton P, Rocella EJ, et al. Prevalence of hypertension in the US adult population. Results from the Third National Health and Nutrition Examination Survey, 1988-1991. *Hypertension*. 1995;25:305-313.)

hypercalcemia have underlying primary hyperparathyroidism.⁴⁹ Hypocalcemia is less common and is observed mainly in patients with advanced CKD (in association with vitamin D deficiency and hyperphosphatemia), chronic malabsorption, and severe malnutrition. Hypomagnesemia is reported in 7% to 10% of elderly patients admitted to the hospital, most commonly because of malnutrition or laxative or diuretic use. Hypermagnesemia is less common and is found primarily in patients with CKD or who are taking large doses of magnesium-containing antacids. Gout (as well as asymptomatic elevations in serum uric acid levels) is also more common in older people.

ENDOCRINE FUNCTION AND KIDNEY HORMONES

Elderly females with eGFR less than 60 mL/min/1.73m² have lower calcium absorption and lower 1,25-hydroxyvitamin D levels, probably because of diminished conversion of 25-hydroxyvitamin D to 1,25-dihydroxyvitamin D by the aging kidney.⁵⁰

The kidney removes about 50% of insulin in the peripheral circulation by filtration and proximal tubular uptake and degradation. The decline of kidney function in the elderly leads to a decrease in insulin clearance. This is, in part, offset by diminished glucose tolerance, which may relate to the increasing frequency of obesity observed in aging individuals. However, the risk for hypoglycemia related to insulin use is increased in the elderly.

CLINICAL MANIFESTATIONS

General Considerations

Aging is associated with declines in kidney function that ultimately limit the ability to defend against destabilizing events. Moderate fluid loss (e.g., an episode of diarrhea) and moderate fluid loading (e.g., inappropriate perioperative intravenous fluids) may be poorly tolerated and lead to hypovolemia and fluid overload, respectively. Hypovolemia in those taking multiple medications with impact on kidney hemodynamics, including angiotensin-converting enzyme (ACE) inhibitors, angiotensin receptor blockers (ARBs), and nonsteroidal anti-inflammatory drugs (NSAIDs), may lead to AKI. Overzealous administration of water as 5% dextrose or 0.45% saline can result in hyponatremia, especially in patients taking selective serotonin reuptake inhibitors, which may increase levels of antidiuretic hormone. The use of NSAIDs in the elderly is associated with increased risk for hyponatremia, hyperkalemia, hypertension, and impaired kidney function.

Glomerular Diseases

Elderly patients may have treatable kidney diseases identifiable on kidney biopsy.⁵¹ In a study of 235 biopsy specimens in patients older than 80 years, 67% of the patients had treatable lesions.⁵² The pathologic spectrum of glomerular disease seen in elderly people is similar to that in the general population, although the prevalence of various pathologies differs. For example, diabetic kidney disease is seen with increasing frequency in the aging population. Among patients with nephrotic syndrome who are older than 60 years, membranous nephropathy is the most common diagnosis, followed by amyloidosis (typically light chain–derived) and minimal change disease (Fig. 93.8).⁵² Other important causes of glomerular disease in the elderly include rapidly progressive glomerulonephritis (GN) resulting from pauci-immune (antineutrophil cytoplasmic antibody-associated) GN (accounting for about 30% of elderly patients with AKI who undergo kidney biopsy).⁵¹ A recently described entity, immunoglobulin G4 (IgG4)-related kidney disease is seen with a higher frequency in males older than 65 years.⁵³ In contrast, certain glomerular disorders, such as lupus nephritis and IgA nephropathy, are uncommon in

Epidemiology of Biopsy-Proven Primary Glomerulonephritis by Age

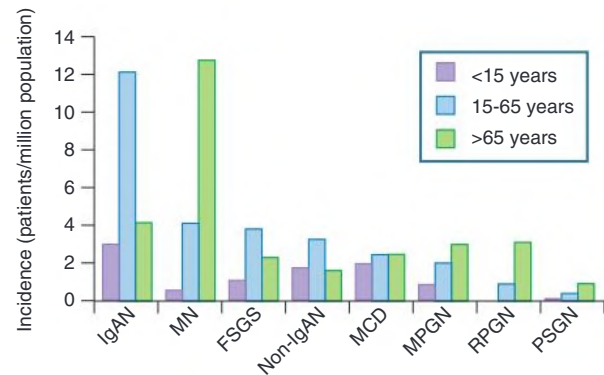


Fig. 93.8 Epidemiology of Biopsy-Proven Primary Glomerulonephritis by Age. Children presenting with nephrotic syndrome are often treated empirically for minimal change disease (MCD); thus diagnosed MCD is likely underrepresented in data derived from biopsy registries. FSGS, Focal segmental glomerulosclerosis; IgAN, immunoglobulin A nephropathy; MPGN, membranoproliferative glomerulonephritis; non-IgAN, other mesangial proliferative glomerulonephritis; PSGN, poststreptococcal glomerulonephritis; RPGN, rapidly progressive glomerulonephritis. (Modified from Vendemia F, Gesualdo L, Schena FP, D'Amico G. Epidemiology of primary glomerulonephritis in the elderly. Report of the Italian Registry of Renal Biopsy. *J Nephrol.* 2001;14:340–352.)

elderly people. Only 2% of patients with lupus nephritis present after the age of 60 years.⁵²

Renovascular and Atheroembolic Disease

There is an increased frequency of renovascular and atheroembolic disease with aging. In several case series, AKI resulting from atheroembolic disease accounted for 4% to 7% of cases.⁵⁴ Atherosclerotic renal artery stenosis is estimated to be present in about 7% of patients older than 65 years and is a major cause of secondary hypertension, ischemic nephropathy, and CKD in the elderly.⁵⁴ In elderly people with hypertension, elevated serum creatinine, and a history of vascular disease, testing for renovascular disease with magnetic resonance angiography or renal artery duplex scanning may be considered (see Chapter 43). Percutaneous transluminal renal angioplasty and renal artery stenting are of variable value in the elderly because many patients have significant arteriosclerosis, which may limit the benefits of intervention.⁵⁴ Thus, an individualized approach to therapy is warranted that weights the likelihood of improvement in GFR and BP against the potential risks of the procedure.

Diabetic Kidney Disease

Diabetic kidney disease (DKD) is a common disease in the elderly; more than half of all diabetic individuals in the United States are older than 60 years. In type 2 diabetic patients, the prevalence of both eGFR decline and albuminuria increases with age.⁵⁵ In patients older than 65 years with type 2 DM, 44.3% had eGFR less than 60 mL/min/1.73 m² and 38.7% had albuminuria, with 17.8% having both findings.⁵⁵ Relatively healthy older adults with no major comorbidities may benefit from more intense glucose control (target hemoglobin [Hb] A_{1c} < 7%), whereas more lenient targets may be more appropriate for elderly with major comorbidities, established diabetic end-organ damage, or limited life expectancy. The doses of oral hypoglycemic agents and insulin may need to be decreased as the kidney function declines,

especially in the elderly, to avoid hypoglycemia and other side effects.⁵⁶ Generally, specific guidelines for the treatment of DM in older people address in a very limited manner the use of more recent therapies, such as sodium-glucose cotransporter-2 inhibitors (SGLT2i), which may have important benefits for older people, such as a low risk of hypoglycemia and reduction of cardiovascular and kidney disease risk. The use of SGLT2i in the elderly must take into account the use of other hypoglycemic medications and drugs such as loop diuretics that may negatively impact volume status.⁵⁷ Kidney function should be monitored closely when introducing these agents.

Acute Kidney Injury

Polypharmacy associated with aging greatly increases susceptibility to the development of AKI as a result of drug toxicity.⁵⁸ Although the causes of AKI in the elderly patient encompass the same spectrum of prerenal, renal, and postrenal causes that are seen in other age groups, the elderly patient has a higher relative risk for developing AKI from obstructive uropathy or sepsis and associated with cancer and anti-neoplastic therapies. Decision making about renal replacement therapy (RRT) in elderly patients with AKI and multisystem organ failure will be influenced by comorbidity and the expected likelihood of a good clinical outcome. These discussions require a multidisciplinary approach.

Nephrotoxicity and Drug Dosage

Elderly patients are prone to increased nephrotoxicity because they are often administered medicines on the assumption that normal or nearly normal serum creatinine concentration is consistent with normal kidney function. Thus, eGFR or perhaps creatinine clearance (but not serum creatinine) should be used to determine drug doses. In addition, elderly patients with CKD usually are prescribed multiple medications, and this creates a great risk for drug-drug interactions that can be exacerbated when GFR is low.

End-Stage Kidney Disease and Kidney Replacement Therapy

The mean age for a patient to initiate RRT is currently over 60 years in the United States and Europe. The mean age of patients receiving maintenance dialysis in most low- and middle-income countries is much lower (32–42 years) than that in high-income countries. In the United States there has been a slight decline in the incidence of ESKD in patients older than 65 years, with a continued higher but stable incidence in those older than 75 years.⁵⁹ However, the prevalence of RRT continues to rise in patients older than 65 years (Fig. 93.9). Despite the frequency of CKD among elderly patients, kidney replacement therapy is far less common as an outcome than events associated with cardiovascular morbidity or mortality. For example, older patients with CKD stage 3 are more likely to die of cardiovascular disease and less likely to receive RRT than their younger counterparts.⁶⁰

The decision to offer RRT should not be based solely on the age of the individual. It is important to recognize that such a decision involving an elderly patient is more complex and fraught with more challenges than in younger patients and requires a multidisciplinary approach that involves family members. Nonmedical barriers, such as limited transportation, family support, and cost, may be particularly important in the elderly. Of note, nursing home patients who began dialysis experienced sharp and sustained declines in their ability to perform basic daily activities.⁶¹ A year after beginning dialysis, 58% of the patients had died and only 13% still functioned at the same level as they had before beginning dialysis. Thus, although dialysis may extend life, it does not necessarily lead to improvements in functional status in older patients. In fact, the increased longevity associated with institution of dialysis may be entirely accounted for by increased days in the hospital without improvements in quality of life (QOL).⁶²

As with other populations, the survival of arteriovenous fistulas (AVFs) in elderly patients is significantly greater than that of arteriovenous grafts.⁶³

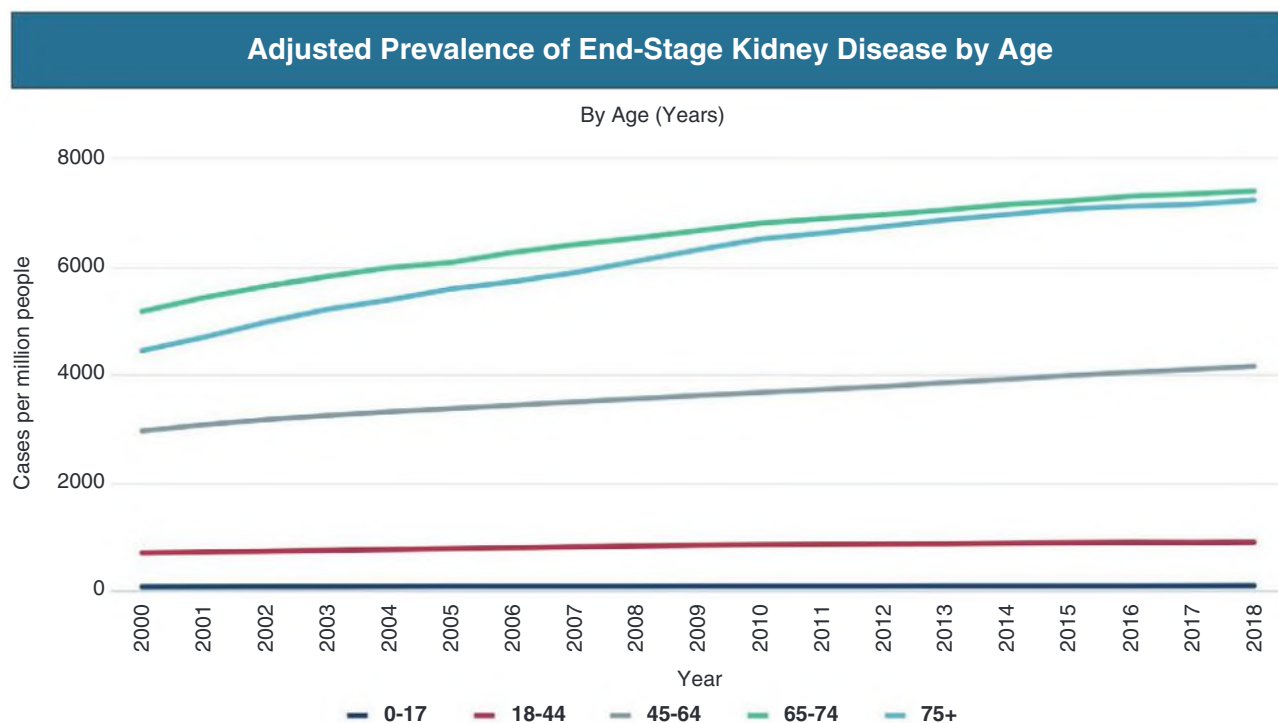


Fig. 93.9 Adjusted Prevalence of End-Stage Kidney Disease (ESKD) by Age. Adjusted ESKD prevalence increased with advancing age, with 7401 cases per million people aged 65 to 74 years and 7233 cases per million people age 75 years or older. (Data from U.S. Renal Data System. www.usrds.org.)

In fact, successful use of an AVF in the elderly has results similar to those in younger people, with prolonged patency and low incidence of infections and thromboses, and thus age alone should not preclude AVF creation. Similarly, use of tunneled hemodialysis (HD) catheters is also associated with increased mortality in older HD patients.⁶³ However, the high prevalence of comorbidities, particularly DM, peripheral vascular disease, and congestive heart failure, usually make vascular access creation more difficult in the elderly. Furthermore, many of these patients may have an insufficient vasculature for fistula maturation. Finally, many fistulas may never be used because of the competing risk for death before dialysis initiation. In these cases, an arteriovenous graft or a central venous catheter becomes a valid alternative form of vascular access.

Additional limitations of HD in the elderly may include sensitivity to fluid shifts and the presence of significant cardiac dysfunction, leading to poor tolerance of HD sessions; hence peritoneal dialysis (PD) may theoretically offer some advantage by offering a slow, sustained degree of ultrafiltration that leads to greater hemodynamic stability. Unfortunately, inherent functional limitations may provide impediments to broader use of PD in the elderly, especially in those without social support.⁶⁴ Studies have demonstrated similar outcomes in elderly patients undergoing HD versus PD.⁶⁴ Comprehensive care programs have been designed to support elderly patients to perform PD in their community or at nursing facilities and may offer options to increase the choices that patients have regarding treatment options.⁶⁵

Transplantation should be considered in the management of elderly patients with ESKD because selected elderly transplant recipients have significantly lower mortality compared with otherwise comparable patients who are wait-listed for transplantation.⁶⁶ This survival benefit is most striking for patients with kidney failure caused by DM or hypertension,

but it declines with longer projected waiting times. Even when expanded-criteria donor kidneys are used, a 25% mortality reduction has been shown in those who undergo transplantation compared with those who are wait-listed.⁶⁶ Overall graft survival for the elderly transplant patient is similar to that in younger patients.⁶⁶ It has been argued that lower doses of immunosuppression are sufficient in elderly kidney transplant recipients. However, the Eurotransplant Senior Program, allocating kidneys from donors 65 years or older to recipients 65 years or older regardless of human leukocyte antigen (HLA) matching, found 5% to 10% increased rejection rates in this “old to old” group compared with two better HLA-matched groups, the “old to any” and the “any to old,” within the Eurotransplant Kidney Allocation System.⁶⁷ In a subanalysis of the Efficacy Limiting Toxicity Elimination (ELITE)-Symphony study, a large prospective randomized trial comparing immunosuppressive regimens in kidney transplant patients, equal rates of rejection in elderly (≥ 60 years) versus younger (< 60 years) recipients were demonstrated.⁶⁸ Furthermore, this subanalysis suggested that older recipients who receive a marginal kidney from an older or expanded-criteria donor may fare worse than younger recipients, with an increased likelihood of death, delayed graft function, graft loss, treatment failure, and reduced graft function.

For those who decide not to pursue RRT, symptomatic therapy (supportive management and palliative care) is an option. This concept underscores the importance of individualizing patient goals with a focus on QOL. For those with severe functional or cognitive impairment or for whom complications of dialysis have negatively affected their QOL, dialysis discontinuation is an important consideration. Hospice services remain underused for ESKD patients but may provide an important support mechanism for patients and their families who wish to discontinue dialysis.

SELF-ASSESSMENT QUESTIONS

1. An 82-year-old man has a sudden onset of severe lower extremity edema and weight gain (6 kg in 10 days). He has been taking allopurinol for gout prophylaxis, celecoxib for pain associated with osteoarthritis, losartan for hypertension, and metformin for type 2 diabetes. He has smoked half a pack of cigarettes daily for the past 60 years. His BP is 160/90 mm Hg. Other than the pitting lower extremity edema, the physical examination findings, including findings from a retinal examination, are within normal limits. Chest radiograph shows mild emphysematous changes and small bilateral pleural effusions. A urinalysis reveals 4+ protein and 1+ blood. Serum creatinine is 3.6 mg/dL, and serum albumin is 2.1 g/dL. The erythrocyte sedimentation rate is 90 mm/h (Westergren). Blood glucose is 140 mg/dL (nonfasting). Electrolytes show sodium of 132 mEq/L, potassium of 5.5 mEq/L, bicarbonate of 20 mEq/L, and chloride of 98 mEq/L. A kidney biopsy is performed. Complete blood count is within normal limits. Which one of the following lesions is *most* likely to be present in the kidney biopsy?
 - A. Minimal change disease (MCD) with interstitial nephritis
 - B. Amyloidosis
 - C. Crescentic glomerulonephritis
 - D. Membranous nephropathy
 - E. Immunoglobulin A nephropathy
2. An 87-year-old woman with hypertensive nephrosclerosis was started on HD 3 months ago. Before dialysis initiation, she had significant weight loss and anorexia, with a serum albumin of 2.8 g/dL. Most recently, her appetite has improved and with this her albumin has improved to 3.2 g/dL, with a serum phosphorus of 5.4 mg/dL and a serum calcium of 9.6 mg/dL. She is currently living in a nursing facility and unable to ambulate without assistance. She appears depressed and apathetic and reports that she has little energy and motivation to participate in any exercise. The family has asked you to meet with them to tell them what they can expect for their mother's health in the future. Which statement is *correct*?
 - A. Dialysis will likely continue to improve her energy level and appetite; the family should expect continued improvement over the next few months.
 - B. Her inability to ambulate, frailty, and depression will likely not improve with dialysis, and her condition is likely to continue to deteriorate over time.
 - C. Home HD should be considered because this modality is associated with improved outcomes in this age group.
 - D. Assisted PD should be considered as a way to keep the patient at home; this modality is associated with increased survival over in-center HD.
3. A 79-year-old patient who recently started HD for ESKD secondary to drug-induced interstitial nephritis approaches you to discuss his potential for kidney transplantation. His clinical history includes hypertension, mild angina-like symptoms, and hyperlipidemia. He does not smoke and is very active. In your region, the average time for patients with his blood type to be on the wait list is 5 years before transplantation. Which of the following statements would *most* accurately inform the patient with regard to his candidacy for kidney transplantation?
 - A. He is unlikely to gain any benefit from transplantation because there is a high risk for death in the first year postoperatively in patients older than 75 years.
 - B. He would likely benefit from transplantation, and he should consider living donation as the best option.
 - C. He should seek transplant listing but not seek a living donor.
 - D. He should be advised to avoid going on the extended-criteria donor waiting list.